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Teaching data science workshops in academic libraries: Insights from attendance patterns and topic preferences

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ABSTRACT

As data science becomes increasingly essential across academic disciplines, academic libraries have emerged as key providers of data literacy and research computing training. This study analyzes attendance data from data science workshops offered by our university libraries for graduate students to understand participation patterns, topic preferences, and audience needs. Results show that workshops focused on data visualization and introductory machine learning attracted the highest attendance across both STEM and non-STEM fields. While advanced machine learning sessions had smaller size attendance, they drew sustained interest from research-oriented participants such as PhD students and postdoctoral researchers. Attendance in programming workshops varied by language, with R and Python emerging as the most popular options. These findings highlight the importance of maintaining a balanced workshop portfolio that supports both foundational and advanced learners. The analysis also reveals opportunities to improve accessibility through standalone design and flexible delivery formats that accommodate diverse schedules and learning backgrounds. Overall, this study provides actionable insights into how academic libraries can better support graduate-level data science education and serve as a model for similar initiatives at other institutions.

Introduction

With the rapid growth of big data and the increasing reliance on data-driven decisions and AI technologies crossing the fields, data science has become an essential discipline in higher education (McFarland et al., 2021; Meng, 2025). To meet the surging demand for data science expertise in both academia and industry, many colleges and universities now offer a wide range of data science courses, degree programs, and certificate options to prepare students for a data-centric world (Acm Data Science Task Force, 2021; Hagen, 2020; Johnson et al., 2024). While these initiatives often provide numerous learning opportunities for undergraduate students (Dogucu et al., 2025; Guzman et al., 2019), data science resources for graduate students, particularly for those outside such fields as computer science, statistics, and business, can still be limited.

This limitation arises in part from the specialized domain knowledge required at the graduate level (University of Kansas et al., 2015). Graduate students are often engaged in advanced research that relies on dedicated data analysis methods and subject-specific tools, resulting in curricula that are highly tailored to individual disciplines (Velaj et al., 2023). Faculty and curriculum developers often face challenges and heavy workload when tasked with redesigning or adapting data science components into these specialized graduate courses, which can hinder the development of domain-specific data science instructions (Bach et al., 2023; Brownell & Tanner, 2012). In addition, foundational data science courses geared towards graduate students are often limited, making it difficult for students to acquire essential data skills at a more advanced level (Yang et al., 2025). As a result, observations on our campus indicate that many graduate students are compelled to enroll in upper-level undergraduate courses (300 or 400 level courses) in order to gain the necessary data science background (Baumer, 2015), highlighting a significant gap in graduate-level offerings.

To help bridge this gap in graduate education, academic libraries and graduate schools at our institute are increasingly working together to offer data science workshops tailored to the needs of graduate students from a broad range of disciplines (Hazzan, 2023). These workshops provide practical, hands-on experience with essential data science tools and methodologies, enabling students to integrate data-driven techniques into their specialized research areas. While such workshops and short-format training sessions can give graduate students valuable initial exposure to key concepts, research shows that their impact on long-term skill retention is often limited unless supplemented by thoughtfully

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